

Tri-Mic Specifications

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Revision History

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Specifications

1 SCOPE

(a) **General:** This document describes the requirements and verification processes for the Tri-Mic. The Tri-Mic is a 3D direction-finding microphone that uses three mics in a triangular array to figure out where sounds are coming from. It runs the MUSIC algorithm to pinpoint direction in three dimensions. The device is built for field technicians and hardware people who need to locate sound sources without moving to a separate workstation. The user can steer the directional focus with a physical control and see the results on a built-in spectrogram display. Captured data can be exported to a PC for further analysis.

(b) **Acronyms:**

- ADC – Analog-to-Digital Converter
- DSP – Digital Signal Processing
- LCD – Liquid Crystal Display
- MUSIC – Multiple Signal Classification
- PCB – Printed Circuit Board
- TDOA – Time Difference of Arrival

2 APPLICABLE DOCUMENTS

The following documents form part of the specifications for this project. In the event of a conflict between requirements, priority goes first to the contract, second to this document, and lastly to these reference documents.

(a) **Government Documents**

1. MIL-STD-810H, Environmental Engineering Considerations and Laboratory Tests, 2019.

(b) **Industry Documents**

1. IEC 61672-1:2013, Electroacoustics – Sound Level Meters.
2. IEC 60529:2004, Degrees of Protection Provided by Enclosures (IP Code).
3. R. Schmidt, "Multiple Emitter Location and Signal Parameter Estimation," IEEE Trans. Antennas Propag., vol. AP-34, no. 3, March 1986.

3 STAKEHOLDER REQUIREMENTS

The stakeholders for the Tri-Mic are:

1. End users (field technicians and hardware enthusiasts)
2. Environmental and durability concerns
3. Data and export needs
4. Manufacturing and engineering team

3.1 Stakeholder User Stories

- **The end user operates the Tri-Mic in the field. The device:**

    1. Must pinpoint specific sounds in 3D and isolate them from background noise.

        User Story
        I work around loud equipment and need to figure out which direction a problem sound is coming from. Right now I just use my ears and walk around until I think I found it. The mic should tell me the direction so I'm not guessing.

    2. Must show a spectrogram on the device without needing a laptop.

        User Story
        I don't want to carry a laptop out to every job. If I can see the frequency data right on the device I can make decisions on site and keep working.

    3. Must let the user steer the directional focus and see the spectrogram change in real time.

        User Story
        I want a knob or joystick that I can move around and watch the display update as I sweep different directions. Something physical makes the most sense—I might have gloves on or be in a tight spot where a touchscreen isn't going to work.

    4. Must be small enough to fit into tight spaces.

        User Story
        Half the places I need to check are behind panels or inside enclosures. The mic needs to get in there without me having to shove my head in.

- **The device must handle real-world field conditions:**

    1. Must be tough enough to handle drops and field use.

        User Story
        Stuff gets dropped. If this breaks from falling off a ladder it's not going to work for us.

    2. Must still work in damp conditions.

        User Story
        I work in crawl spaces, mechanical rooms, and sometimes outside. A little moisture or humidity shouldn't kill it.

- **Captured data needs to be useful after the fact:**

    1. Must be able to export audio and directional data to a PC.

        User Story
        The on-device display is fine for field work but for real analysis we need to pull the data off and run it through our own tools on a computer. We want WAV files and TDOA data we can actually work with.

- **Manufacturing and engineering cares about cost and flexibility:**

    1. Must use standard off-the-shelf components.

        User Story
        We don't want exotic parts with long lead times. Standard mics, standard PCB components, standard connectors. The MUSIC algorithm is well-established so we're not reinventing the wheel on the computation side either.

    2. Must support adjustable mic spacing and swappable DSP pipelines during prototyping.

        User Story
        We need to try different array configurations to see what gives the best direction-finding accuracy. MUSIC is our starting algorithm but we might try other approaches. The architecture should let us change both the mic geometry and the processing without redesigning the hardware.

4 ENGINEERING REQUIREMENTS

4.1 Item Description

The Tri-Mic takes in sound through a triangular mic array, processes it using the MUSIC algorithm, and outputs the direction and frequency content of the sound to the user on a built-in display. The system has two physical parts connected by a boom: a probe and a handheld controller.
The probe holds the three microphones and sits at the end of the boom so it can reach into tight spots. The handheld controller has the display, steering controls, processor, and storage. The user holds the controller and positions the probe where they need to listen.
To do this, the system has five stages: a sound capture module, a signal conditioning module, a processing module, a storage module, and a display module (refer to Figure 1).

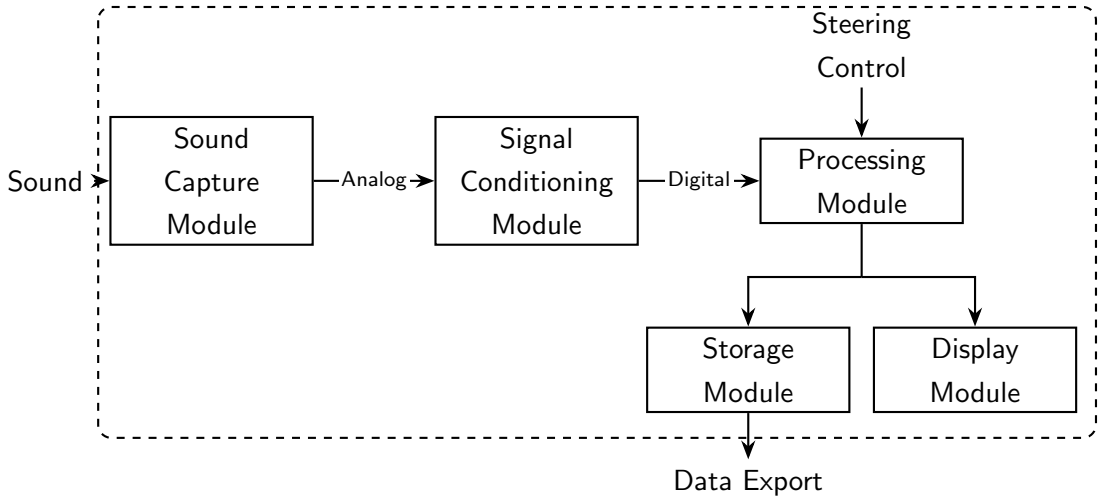


Figure 1: Modularized Functional Diagram

The non-functional parts of the system consist of a power component, physical requirements, and environmental factors (refer to Figure 2).
The following sections describe the requirements of the interface, functional, and non-functional components of the Tri-Mic.

4.2 Interface Requirements

4.2.1 Physical

4.2.1.1 The probe shall have maximum dimensions of 5 cm $\times$ 5 cm $\times$ 5 cm.

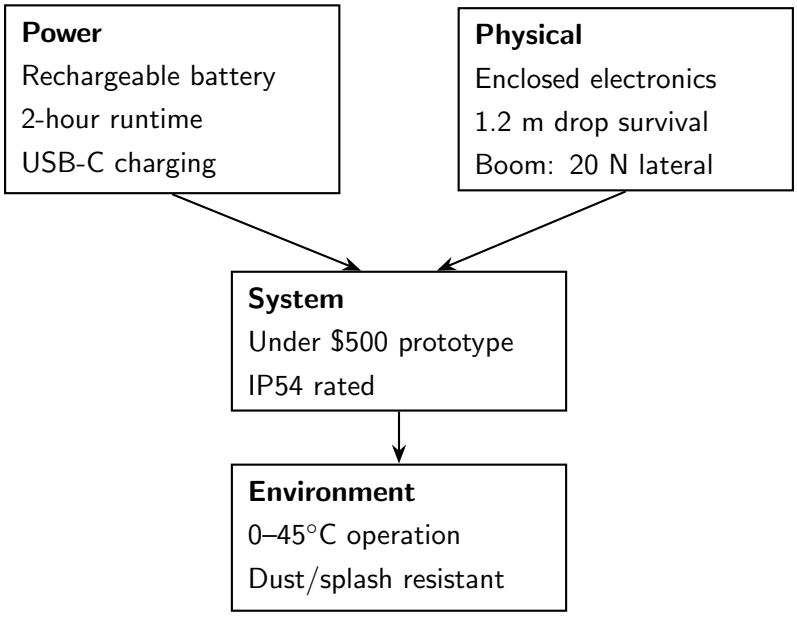


Figure 2: System Functional Support Diagram

4.2.1.2 The boom shall be between 30 cm and 60 cm long. The probe shall attach to the boom with a detachable connector.

4.2.1.3 The handheld controller shall have maximum dimensions of 18 cm $\times$ 10 cm $\times$ 5 cm. The total weight of the assembled system (probe, boom, and controller) shall not exceed 1.5 kg.

4.2.2 Functional

4.2.2.1 The controller shall have a physical steering control (knob, roller, or joystick) for adjusting directional focus, and an LCD display with at least 320 $\times$ 240 resolution.

4.2.2.2 The system shall have a data port (USB or similar) for exporting data to a PC.

4.3 Functional Requirements

4.3.1 Sound Capture Module

4.3.1.1 The system shall use three microphones arranged in a triangular array, capturing frequencies from 20 Hz to 20 kHz.

4.3.1.2 Each mic channel shall have a signal-to-noise ratio of at least 60 dB.

4.3.2 Signal Conditioning Module

4.3.2.1 The signal conditioning module shall amplify each mic output to a level that works with the ADC input range.

4.3.2.2 The ADC shall sample all three channels simultaneously at a minimum of 44.1 kHz with at least 16-bit resolution.

4.3.3 Processing Module

4.3.3.1 The processing module shall run the MUSIC algorithm for direction-of-arrival estimation. The system shall determine azimuth and elevation of a sound source within $\pm 5^{\circ}$ at distances of 1 m or more.

4.3.3.2 The processing module shall do beamforming so the user can focus on one direction and reject sound from other directions. The system shall isolate a target sound source with at least 15 dB of directional rejection.

4.3.3.3 The system shall generate a 2D time-by-frequency spectrogram from the beamformed output, updating at least 10 times per second.

4.3.4 Storage Module

4.3.4.1 The system shall store captured audio and directional data on at least 1 GB of onboard non-volatile memory.

4.3.4.2 The system shall export raw audio as WAV files and directional data as CSV files.

4.3.5 Display Module

4.3.5.1 The display shall show the spectrogram with a directional indicator and a 3D arrow or similar visual pointing toward the dominant sound source.

4.3.5.2 Latency from steering input to display update shall be no more than 100 ms.

4.4 Non-Functional Requirements

4.4.1 Electrical

4.4.1.1 The system shall run on a rechargeable battery that lasts at least 2 hours of continuous use.

4.4.1.2 The system shall have a USB-C charging port.

4.4.2 Physical

4.4.2.1 All electronics shall be enclosed in a protective case. The system shall survive a 1.2 m drop onto concrete without breaking.

4.4.2.2 The boom shall handle at least 20 N of lateral force without bending permanently.

4.4.3 Environmental

4.4.3.1 The system shall operate from 0°C to 45°C.

4.4.3.2 The system shall be at least IP54 rated (dust protected, splash resistant).

4.4.4 Economic

4.4.4.1 The prototype shall cost no more than $500 in components.

4.5 Other Requirements

4.5.1 Safety

4.5.1.1 The system shall not have exposed conductors during normal use. The system shall not produce sound louder than 85 dBA at the user's ear.

4.5.1.2 The battery shall have over-current and over-temperature protection.

5 VERIFICATION OF REQUIREMENTS

Possible verification methods include:

1. Inspection:
   Visual examination, measurement, or gauging without special lab equipment.
2. Demonstration:
   Showing that the system works by operating it. No special instrumentation needed.
3. Analysis:
   Review of documentation, calculations, or accumulated test data.
4. Direct Test:
   Using special equipment or instrumentation to measure specific performance characteristics.

5.1 Interface Requirements

5.1.1 Physical

5.1.1.1 Measure the probe with calipers. Verify it fits within 5 cm $\times$ 5 cm $\times$ 5 cm.

5.1.1.2 Measure the boom length. Verify it's between 30 cm and 60 cm. Disconnect and reconnect the probe 10 times. Verify the connection holds and the system still works.

5.1.1.3 Measure the handheld controller. Verify it fits within 18 cm $\times$ 10 cm $\times$ 5 cm. Weigh the full system on a scale. Verify total is 1.5 kg or less.

5.1.2 Functional

5.1.2.1 Turn the steering control. Verify the beamforming direction changes on the display. Check the LCD datasheet. Verify resolution is at least 320 $\times$ 240.

5.1.2.2 Plug the system into a PC via the data port. Verify the PC sees the device and data transfers.

5.2 Functional Requirements

5.2.1 Sound Capture Module

5.2.1.1 Inspect the mic array. Verify three mics are arranged in a triangle. Play test tones at 20 Hz and 20 kHz with a calibrated speaker. Record and verify both frequencies show up in the data.

5.2.1.2 Measure SNR on each channel using a 1 kHz tone at 94 dB SPL. Verify all channels hit at least 60 dB.

5.2.2 Signal Conditioning Module

5.2.2.1 Feed a known signal into the preamp. Verify the output level is in the ADC's input range.

5.2.2.2 Record a test signal and check the file properties. Verify sample rate is at least 44.1 kHz at 16-bit or better. Clap near the array. Check the recorded arrival times on all three channels. Verify the offsets match the known mic spacing, confirming the channels are in sync.

5.2.3 Processing Module

5.2.3.1 Place a sound source at a known position 2 m away. Run the system and verify the estimated direction is within $\pm 5^{\circ}$ of actual. Repeat at 5 different angles.

5.2.3.2 Put two sound sources at different directions. Steer toward one and verify the other is at least 15 dB quieter in the output.

5.2.3.3 Watch the spectrogram for 10 seconds. Count the frame updates. Verify at least 10 frames per second.

5.2.4 Storage Module

5.2.4.1 Check the storage chip datasheet. Verify at least 1 GB capacity.

5.2.4.2 Record audio and export. Open the WAV and CSV files on a PC and verify they contain valid data.

5.2.5 Display Module

5.2.5.1 Run the system with a sound source at a known direction. Verify the display shows a spectrogram with a directional indicator and a 3D arrow that matches the actual direction. Move the source. Verify the arrow updates to follow it.

5.2.5.2 Turn the steering control and time how long until the display updates. Verify latency is under 100 ms.

5.3 Non-Functional Requirements

5.3.1 Electrical

5.3.1.1 Inspect the system. Verify it runs on a rechargeable battery. Charge fully and run continuously. Time it until it dies. Verify at least 2 hours.

5.3.1.2 Inspect for a USB-C port. Plug in a charger and verify it charges.

5.3.2 Physical

5.3.2.1 Inspect the system. Verify all electronics are enclosed. Drop the controller and probe each from 1.2 m onto concrete. Power on after each drop and verify everything still works.

5.3.2.2 Push on the boom sideways with a force gauge at 20 N. Let go and verify it didn't bend.

5.3.3 Environmental

5.3.3.1 Run the system at 0°C and 45°C. Verify it works at both extremes.

5.3.3.2 Run IP54 testing per IEC 60529 (dust and splash). Verify the system still works after.

5.3.4 Economic

5.3.4.1 Add up the BOM. Verify total component cost is $500 or less.

5.4 Other Requirements

5.4.1 Safety

5.4.1.1 Inspect the system. Verify no conductors are exposed during normal use. Measure sound output at the user's ear with a sound level meter. Verify it doesn't exceed 85 dBA.

5.4.1.2 Check the battery protection circuit datasheet. Verify it has over-current and over-temperature protection.

5.5 Verification Summary

Req.	Test Type	Tester's Name	Pass/Fail	Date
4.2.1.1	Inspection			
4.2.1.2	Inspection			
4.2.1.3	Inspection			
4.2.2.1	Demonstration			
4.2.2.2	Demonstration			
4.3.1.1	Direct Test			
4.3.1.2	Direct Test			
4.3.2.1	Direct Test			
4.3.2.2	Direct Test			
4.3.3.1	Direct Test			
4.3.3.2	Direct Test			
4.3.3.3	Direct Test			
4.3.4.1	Analysis			
4.3.4.2	Demonstration			
4.3.5.1	Demonstration			
4.3.5.2	Direct Test			
4.4.1.1	Direct Test			
4.4.1.2	Inspection			
4.4.2.1	Direct Test			
4.4.2.2	Direct Test			
4.4.3.1	Direct Test			
4.4.3.2	Direct Test			
4.4.4.1	Analysis			
4.5.1.1	Inspection			
4.5.1.2	Analysis			